Four questions your table can answer

TC1037 · Algorithmic Thinking · Session 14 · Deliverable 2, written here

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Dr. Leopoldo Rodríguez Salazar

Friday 13 November 2026

Tecnológico de Monterrey · Campus Querétaro · Group 510

Tonight at 23:59

Deliverable 2 is due in eight hours.

What it is. Four to six *preguntas detonadoras* about *your* table. One Word file per team, `entrega2.docx`.

It is not code. Nobody has to open Python tonight.

What we are doing about it.

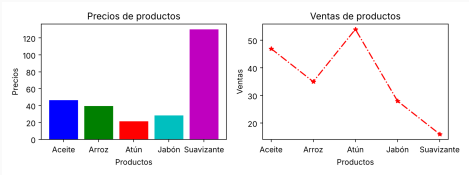
Twenty minutes of this session are yours, in pairs, to write it. Most of you will leave this room with the file finished.

Bring it up on a laptop now if you have one.

You just spent an hour on this in the lab

In Luis's hour you wrote three functions over two short lists:

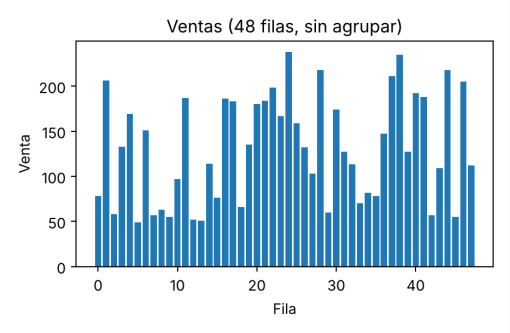
`precios_productos` — a bar chart
`ventas_productos` — a line chart
`precios_ventas` — both, side by side

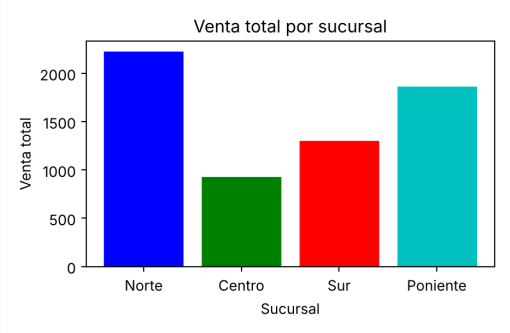


Every one of them took the same two things: **a list of labels** and **a list of numbers, the same length.**

The lab handed you those two lists already made. **Your project will not.**

Where the work actually is





Your forty rows, drawn directly. True, complete, useless.

The same rows, **grouped first**. Now it answers something.

A chart does not come from a table. It comes from a question asked of a table.

Which is why the questions are due before the charts.

Deliverable 2 — the exact shape

Four to six questions. Each one carries three things, and then an owner.

	Write	Meaning
1	The **question**	One a manager would ask out loud. *Which branch sells most?* — not *what is in the data?*
2	The **column or columns**	By their real names, spelled as they are in your file.
3	The **statistic or chart**	*Total by branch, bar chart. Mean price, one number.*
4	The **owner**	Each of you takes two or three, written down.

Marks: **50** questions answerable from your columns · **30** naming the column and the statistic or chart · **20** the ownership.

Worked, on tables that are actually in this room

Somebody's table	A question it can answer	Columns	Statistic / chart
Car sales by model and year	Which model holds its price best over the years?	`Model`, `Year`, `Price`	Mean price by model · line chart
Café nutrition by product	Are the bakery items heavier in sugar than the cookies?	`Category`, `Sugar`	Mean by category · bar chart
Museum visitors by venue	Which three venues bring in the most visitors?	`Venue`, `Visitors`	Total by venue · bar chart
Purchase orders by supplier	Which supplier costs us most per order?	`Proveedor`, `Importe`	Mean by supplier · bar chart
Health indicators by country	Do higher-income regions live longer?	`Income Group`, `Life expectancy`	Mean by income group · bar chart

Every one of these is **group a text column, measure a numeric column**. That one sentence is four of your six questions.

Three tests before you write a question down

Can I name the column?	If no, your table cannot answer it. Change the question, not the table.
Is the answer a number or a picture?	If the answer is a sentence, it is an opinion, not a result.
Would anyone act on it?	If the answer changes nothing, it is trivia.

Four to six questions. If you have three good ones and three weak ones, hand in the three good ones and one more you believe in.

Four that fail, and the repair

Written as	Why it fails	Repaired
Are our customers satisfied?	No column holds it	Which product is returned most often? (`Producto`, `Devuelto`)
What is in the data?	Not a question	Which category has the highest mean price?
How many rows are there?	Nobody acts on it	Which month had the lowest total sales?
Is the URL correct?	That column is a label, not a variable	Drop the column; cite the source in the note

Row one is a good business question and a bad deliverable-2 question. The difference is the only thing being graded.

Now. Twenty minutes, in pairs.

Open your `entrega1.xlsx` and read your own column names out loud to each other. That alone produces two questions.

Write four. For each: the question, the columns, the statistic or chart.

Then split them — two or three each — and **write the names down in the file**.

I am walking around.

Stop me when you are unsure whether a question is answerable. That is a ten-second conversation and it is exactly what I am here for.

No team of one tonight — if your partner is absent, work with the pair next to you and split the file afterwards.

At the end of the twenty minutes I will ask three teams to read one question out. Not to judge it — to show the rest what one sounds like.

Before you upload tonight — the checklist

- ☐ Four to six questions, each one a sentence a manager would say
- ☐ Every question names its **column or columns**, spelled as in the file
- ☐ Every question names a **statistic or a chart**
- ☐ **Both names** on the file, and who owns which questions
- ☐ Saved as `entrega2.docx` — a Word file, not a PDF, not a photo
- ☐ Uploaded to **Canvas**, by **one** of you, before 23:59

One upload per team. Not by email.

And while you are in there — Deliverable 1

Most of you have a message from me with a list of about three lines.

Blank rows and columns · a column with the same value in every row · an identifier used as a variable · numbers stored as text.

Fixing it carries no penalty. Upload the corrected file whenever it is ready.

Why it matters more next week than it did last week.

On **20 November** your program opens this file with `pandas` and reads it.

A blank row becomes a row. A number stored as text stops being a number. Neither raises an error — you just get a wrong answer that looks reasonable.

What happens to this file next

When	What	
Tonight	Deliverable 2	The questions.
Fri 20 Nov	Deliverable 3	`entrega3.py` — your Excel loaded into a DataFrame and printed.
Fri 27 Nov	Final, **individual**	The program answers your questions, with the two charts and the report.
Tue 1 Dec	Final exam	

The questions you write tonight are the ones your program answers in two weeks. Choose ones you would like to know the answer to.

Before you go

A question you cannot trace to a column is not a question about your data. It is a wish.

Tonight 23:59

`entrega2.docx` · one per team · Canvas

Stuck on Sunday?

Book an asesoría — the link is in the Canvas announcement. Bring the file.